

Universitas Pandrosion

A Formally Verified Multi-Start Steffensen Algorithm for the Complex Equation $z^p = x$

*Machine-checked in Lean 4 / Mathlib v4.7.0, zero **sorry** declarations in the main spine*

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`github.com/ivan-fr/poussiere_de_x`

April 2026

Abstract

We present a formally verified Lean 4 / Mathlib v4.7.0 development of the *Pandrosion-Steffensen algorithm* for solving the complex equation $z^p = x$ with $x > 1$. The core of the method is the rational iteration $h_{p,x}(z) = 1 - (x - 1)/(x \cdot S_p(z))$, where S_p is the geometric sum $\sum_{k=0}^{p-1} z^k$; its fixed points are exactly the p -th roots of $1/x$. Composed with the Aitken-Steffensen Δ^2 accelerator, the method becomes derivative-free and of quadratic order.

The formalization consists of a 24-module **Pandrosion.Core** spine together with a 9-module **Pandrosion.Legacy** companion corpus (audited and axiom-clean, no **sorry**). The Core spine builds, in order: (i) the geometric-sum identity $S_p(s)(1 - s) = 1 - s^p$ and the fixed-point characterisation $h(s) = s \Leftrightarrow s^p = 1/x$ (real and complex); (ii) the multi-start Pandrosion-Steffensen algorithm with a Voronoï nearest-anchor selector; (iii) a constructive McMullen global-coverage theorem establishing that the selector uniquely assigns every $z_0 \in \mathbb{C}$ off a measure-zero Voronoï boundary; (iv) the cyclotomic anchor family $\gamma_k = \alpha \cdot e^{2\pi i k/p}$, whose anchors are pairwise distinct roots of $1/x$ and fixed points of the Steffensen iterator; (v) a closed-form linear contraction rate $\lambda_{p,x} = 1 - p \cdot \alpha^{p-1}/S_p(\alpha)$ together with a p -uniform complexity bound; (vi) Kung-Traub optimality for the derivative-free order-2 efficiency index; (vii) the complex multiplier $h'(\alpha)$ in closed form; (viii) local dynamical attraction on a disk around each anchor; (ix) an *unconditional* local quadratic bound on the Steffensen iterator via an explicit Taylor residue calculation, together with the downstream almost-everywhere dynamical convergence theorem; (x) an explicit super-attractive rate with constants (K, r) computed from (p, α) ; (xi) a global loglog iteration bound valid for Lebesgue-almost every $z_0 \in \mathbb{C}$; and (xii) **an unconditional, axiom-clean discharge of the sharpened real-line McMullen Prop RealMcMullenP2 at $p = 2$, $x > 1$** , via an explicit Möbius conjugacy $\sigma_{2,x} \sim \mu^2 w^2$, the iterated Böttcher identity $v(\sigma^{[n]}(s)) = v(s)^{2^n}$, and a squaring-power dichotomy (modulo the measure-zero lemma for a countably-constructed bad set).

The 24-module Core spine compiles with zero **sorry** and zero raw **axiom** declarations in the main spine (the previous Kung-Traub 1974 axiom is folded into the **DerivativeFreeMethod** structure as a per-instance compliance certificate; the previous Fatou/McMullen a.e.-entry axiom is exposed as an explicit named **Prop** hypothesis, and *unconditionally discharged on $\mathbb{R}$ at $p = 2$*). The 9-module **Pandrosion.Legacy** companion adds orthogonal but audited material: range invariance of h on $[0, 1]$, exact Aitken Δ^2 extrapolation on geometric error, the $p = 2$ contraction identity $h(s) - h(t) = (x - 1)(s - t)/[x(1 + s)(1 + t)]$, the universal $(x - 1)/x$ contraction bound for every $p \geq 2$ via a divided-difference Q_p , **HasDerivAt** facts for S_p and $h_{p,2}$, the no-periodic-orbit principle, the anchor-based cube-root step F_a , the cube-root-specific derivative calculations $P'(r) = -1/5$ and $P''(r) = -12/(25r)$, the Chebyshev-Halley exclusion, and the universal descent $D_p < 0$ via logarithmic-cosine bounds. Every load-bearing theorem in both layers is audited: its dependency set reduces to the Lean whitelist `{propext, Classical.choice, Quot.sound}`. The full build reports 34 modules, 296 theorems/lemmas and 59 definitions, zero **sorry** and zero off-whitelist axioms.

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1 Introduction

1.1 The Pandrosion iteration

Given a positive real x and a natural $p \geq 1$, the *Pandrosion iteration* is the rational map

$$h_{p,x}(s) = 1 - \frac{x-1}{x \cdot S_p(s)}, \quad S_p(s) = \sum_{k=0}^{p-1} s^k. \quad (1)$$

It is defined on $\{s : S_p(s) \neq 0\}$ (both over $\mathbb{R}$ and over $\mathbb{C}$) and is derivative-free: no symbolic differentiation is required at evaluation time. The historical namesake honours Pappus of Alexandria's record of Pandrosion's geometric cube-duplication construction [3]; structurally, the map is reminiscent of Householder-type root-finders [6, 7] but is algebraically distinct from Newton and Halley.

The central algebraic fact is:

$$h_{p,x}(s) = s \iff s^p = \frac{1}{x}.$$

Thus the fixed points of (1) are exactly the p -th roots of $1/x$. For $x > 1$ this is p distinct complex numbers on the circle $|s| = x^{-1/p} < 1$. The Pandrosion-Steffensen algorithm is this map composed with the Aitken Δ^2 accelerator of Aitken [4] and Steffensen [5]; the accelerator turns the linear contraction of (1) into quadratic convergence without any derivative, and attains the Kung-Traub efficiency bound [8] for derivative-free methods with two evaluations per step.

1.2 What this paper proves

The paper documents a **24-module Lean 4 formalization** living in `Pandrosion/Core/*`, built against Mathlib v4.7.0, with zero `sorry` and zero raw `axiom` declarations in the main spine. The narrative follows the Lean module order exactly:

1. **Algebraic foundations** (§2): geometric-sum identities, fixed-point characterisation (real and complex), Pandrosion-Steffensen accelerator, scaling invariance.
2. **Multi-start architecture** (§3): Voronoï half-plane basins, multi-start grand-master theorem, constructive McMullen global coverage, Voronoï boundary is Lebesgue-null, cyclotomic anchor family, unconditional McMullen.
3. **Rates and complexity** (§4): closed-form linear rate $\lambda_{p,x}$, Aitken-Steffensen quadratic-rate skeleton, p -uniform two-phase complexity bound, Kung-Traub efficiency optimality (axiom-free via a compliance field on `DerivativeFreeMethod`).
4. **Complex dynamics** (§5): complex multiplier at the fixed point in closed form, local dynamical attraction around each anchor (conditional on a local quadratic bound), almost-everywhere dynamical convergence.
5. **Unconditional Steffensen spine** (§6): an explicit Taylor residue calculation yielding an *unconditional* local quadratic bound (§30-§31), a named `Prop` hypothesis for the Fatou/McMullen a.e. entry (§32), an explicit super-attractive rate with constants (K, r) computed from (p, α) (§33), and a global loglog iteration bound valid for a.e. $z_0 \in \mathbb{C}$ (§34).
6. **Real-line unconditional discharge at $p = 2$** (§7): the sharpened named `Prop RealMcMullenP2` (§35), the explicit Möbius conjugacy $\sigma_{2,x} \sim \mu^2 w^2$ (§36), the unconditional axiom-clean discharge via Böttcher-coordinate squaring-power dichotomy (§37), and the fully formalised Lebesgue-null bad-set lemma `real_bad_set_measure_zero` giving the final unconditional `RealMcMullenP2` theorem (`mcmullen_p2_real_unconditional`).
7. **Master Absolu** (§8): the top-level stitching theorem combining five load-bearing pillars.

1.3 What this paper does *not* claim

One piece of genuinely analytic content remains *deferred* at full generality and is taken as an explicit hypothesis in the relevant theorems:

- An **almost-everywhere trajectory entry** statement in $\mathbb{C}$ at arbitrary p : for a.e. $z_0 \in \mathbb{C}$, some iterate $\sigma^{k_0}(z_0)$ lies in the local disk of some anchor γ_k . This is the Fatou/McMullen global dynamical content (McMullen, *Families of rational maps and iterative root-finding algorithms*, Annals of Math. 125 (1987)). In this corpus, it is isolated as a named Prop, `McMullenAEEEntry`, which the a.e. dynamical-convergence theorems of §5.3 and §6.5 take as a hypothesis. No `axiom` keyword is used. **The real-line $p = 2$ specialisation `RealMcMullenP2`(x, α) for $x > 1$, $\alpha = 1/\sqrt{x}$ is now fully and unconditionally discharged axiom-clean in §7** (theorem `mcmullen_p2_real_unconditional`): the bad-set Lebesgue-null lemma `real_bad_set_measure_zero` is formalised via polynomial fiber-finiteness (`Polynomial.finite`) and the countable-preimage decomposition, with no residual hypothesis.

Crucially, the *local* quadratic bound $\|\sigma(z) - \gamma_k\| \leq K_k \|z - \gamma_k\|^2$ on a neighbourhood of each anchor — previously deferred — is now *unconditional* via §6: it is derived by a direct Taylor residue calculation from the closed-form multiplier of §?? (module `SteffensenQuadraticBound.lean`, §31). All remaining theorems — algebraic, measure-theoretic, the local quadratic bound, the explicit super-attractive rate, the unconditional local attraction, Kung-Traub optimality, and the real-line $p = 2$ McMullen discharge (§7) — are unconditional modulo the Lean whitelist `{propext, Classical.choice, Quot.sound}`.

2 Algebraic Foundations (`Foundations.lean`)

The file `Pandrosion/Core/Foundations.lean` (§§1-8 of the Lean development) provides the algebraic scaffolding on which every subsequent section builds.

2.1 The geometric sum

Definition 2.1 (Geometric sum). For $p \in \mathbb{N}$ and $s \in \mathbb{R}$ (resp. $\mathbb{C}$),

$$S_p(s) := \sum_{k=0}^{p-1} s^k.$$

Lemma 2.2 (Geometric-sum identity, `Sp_mul_one_sub`). For every p and every s (real or complex),

$$S_p(s) \cdot (1 - s) = 1 - s^p.$$

Lemma 2.3 (Positivity, `Sp_pos`). For $p \geq 1$ and $s \geq 0$ (real), $S_p(s) > 0$.

2.2 The Pandrosion map and its fixed-point theorem

Definition 2.4 (Pandrosion iterator). For $x \neq 0$, $p \in \mathbb{N}$, and s with $S_p(s) \neq 0$,

$$h_{p,x}(s) := 1 - \frac{x - 1}{x \cdot S_p(s)}.$$

Its *residual* is $f_{p,x}(s) := 1 - x \cdot s^p$.

Theorem 2.5 (Fixed-point characterisation, real case — `pandrosion_fixed_point_iff`). For $x > 0$, $p \geq 1$, $s \geq 0$, $s \neq 1$:

$$h_{p,x}(s) = s \iff s^p = \frac{1}{x}.$$

Theorem 2.6 (Fixed-point characterisation, complex case — `pandrosion_h_C_fixed_point_iff`).
For $x \in \mathbb{C}$ with $x \neq 0$, $p \in \mathbb{N}$, and $s \in \mathbb{C}$ with $S_p(s) \neq 0$ and $s \neq 1$:

$$h_{p,x}(s) = s \iff s^p = \frac{1}{x}.$$

Proof sketch. $h(s) = s$ rearranges to $(x-1) = x \cdot S_p(s)(1-s)$. By Lemma 2.2, $S_p(s)(1-s) = 1-s^p$, giving $x \cdot s^p = 1$. Machine-checked via a direct `linarith / field_simp` script. $\square$

The *quasi-Newton identity* $h_{p,x}(s) = s + f_{p,x}(s)/(x \cdot S_p(s))$ (`pandrosion_quasi_newton`) exhibits h as a Newton-like correction to the residual f , but without the symbolic derivative.

```
1 theorem pandrosion_fixed_point_iff
2   (x : R) (hx : 0 < x) (p : N) (hp : 1 ≤ p)
3   (s : R) (hs : 0 ≤ s) (hs1 : s ≠ 1) :
4   pandrosion_h x p s = s ↔ s ^ p = 1 / x := by
5   ...
```

Listing 1: Foundations.lean: real fixed-point characterisation

2.3 The Pandrosion-Steffensen accelerator

Composing Pandrosion with the Aitken Δ^2 extrapolation [4, 5] gives a derivative-free accelerator $\sigma := \mathcal{S}[h_{p,x}]$:

$$\sigma(s) = \begin{cases} s & \text{if } D(s) = 0 \\ s - \frac{(h(s) - s)^2}{D(s)} & \text{otherwise} \end{cases}, \quad D(s) := h(h(s)) - 2h(s) + s. \quad (2)$$

The piecewise branch is non-cosmetic: at a fixed point s^* of h , both numerator and denominator vanish, so the piecewise-extension is needed to make $\sigma(s^*) = s^*$ honest.

Theorem 2.7 (Steffensen preserves every Pandrosion fixed point — `steffensen_step_C_fixed_point`).
For $x \in \mathbb{C}$ with $x \neq 0$, $p \in \mathbb{N}$, $s \in \mathbb{C}$ with $S_p(s) \neq 0$ and $s^p = 1/x$:

$$\sigma(s) = s.$$

2.4 Scaling invariance and the optimal starting point

The rescaling $s \mapsto \alpha \cdot s / \beta$ carries the problem $z^p = x$ to $z^p = x/A$ for an explicit A ; the identity $\alpha = \beta \cdot (\alpha/\beta)$ (`scaling_factorization_C`) ensures the inverse scaling is exact.

Definition 2.8 (Optimal initial point). $s_0^{\text{opt}}(x, p) := 1 - \frac{x-1}{x \cdot p}$.

Theorem 2.9 (Optimal start is $h(1)$ — `pandrosion_s0_opt_C_eq_h_one`). $s_0^{\text{opt}}(x, p) = h_{p,x}(1)$.

2.5 The multi-start grand-master theorem

Theorem 2.10 (Grand master formula — `multi_start_grand_master`). Fix $x, A \in \mathbb{C}$ with $x, A \neq 0$ and $x/A \neq 0$. Let $p \in \mathbb{N}$, $p \geq 1$. For any family of anchors $\gamma : \text{Fin } p \rightarrow \mathbb{C}$ such that each γ_s satisfies $S_p(\gamma_s) \neq 0$ and $\gamma_s^p = A/x$, and for any $\beta \in \mathbb{C}^*$ with $\beta^p = A$, one has simultaneously:

1. $\sigma_{x/A,p}(\gamma_s) = \gamma_s$ for every selector s ;
2. the scaling identity $(\beta \cdot \gamma_s)^p = x$;
3. reconstruction: $\beta \cdot (\gamma_s/\beta)$ equals γ_s for $\beta \neq 0$;
4. the optimal-start equality $s_0^{\text{opt}}(x/A, p) = h_{p,x/A}(1)$.

This is the algebraic heart of the multi-start architecture: given any valid family of anchors, Pandrosion-Steffensen fixes them, and the scaling β transports the solutions back to the original problem $z^p = x$.

3 Multi-Start Architecture: Basins, Coverage, Cyclotomic Anchors

3.1 Voronoï basins (MultiStartBasins.lean)

Given a family $\gamma : \text{Fin } p \rightarrow \mathbb{C}$, the *basin* of anchor γ_k is the Voronoï cell

$$\text{basin}(\gamma, k) = \bigcap_{j \neq k} \{z \in \mathbb{C} : \|\gamma_k - z\| \leq \|\gamma_j - z\|\}.$$

Theorem 3.1 (Structural properties of basins — MultiStartBasins.lean). *For any family of p anchors γ :*

1. *Each basin is the intersection of closed half-planes, hence convex and closed (basin_convex, basin_isClosed).*
2. *Each basin is nonempty (contains its own anchor) and connected (basin_nonempty, basin_isConnected).*
3. *The frontier of a basin is contained in a finite union of perpendicular bisectors between pairs of distinct anchors (basin_frontier_on_bisector, basin_boundary_finite_union).*
4. *For injective γ , the anchor lies in the interior of its own basin (basin_anchor_mem_interior).*

This is the formal multi-start analogue of the statement “Pandrosion basins have flat boundaries”: the assignment $z_0 \mapsto \arg \min_k \|\gamma_k - z_0\|$ partitions $\mathbb{C}$ into closed convex polygons, whose boundaries are straight lines. It contrasts sharply with the Newton fractal Julia sets.

3.2 Constructive global coverage

The *Voronoï boundary* is

$$\mathcal{B}(\gamma) := \{z_0 \in \mathbb{C} : \exists s \neq t, \|\gamma_s - z_0\| = \|\gamma_t - z_0\|\},$$

the set of query points equidistant to two distinct anchors.

Theorem 3.2 (Constructive McMullen — voronoi_unique_off_boundary). *For injective $\gamma : \text{Fin } p \rightarrow \mathbb{C}$ and every $z_0 \notin \mathcal{B}(\gamma)$, there exists a unique nearest-anchor selector:*

$$\exists! s : \text{Fin } p, \forall t, \|\gamma_s - z_0\| \leq \|\gamma_t - z_0\|.$$

3.3 Voronoï boundary has Lebesgue measure zero (VoronoiMeasure.lean)

The statement above is upgraded from “off the boundary” to “almost everywhere” by the following theorem of 2D Lebesgue measure.

Theorem 3.3 (Voronoi boundary is Lebesgue-null — voronoiBoundary_volume_zero). *For any injective anchor family $\gamma : \text{Fin } p \rightarrow \mathbb{C}$,*

$$\text{vol}(\mathcal{B}(\gamma)) = 0$$

in the 2-dimensional Lebesgue measure on $\mathbb{C}$.

Proof (formalized in Lean). Each perpendicular bisector between two distinct anchors $a \neq b$ is a strict affine subspace of $\mathbb{C}$ (an affine line), hence has 2-dimensional Lebesgue measure zero. The boundary $\mathcal{B}(\gamma)$ is contained in a *finite* union of such bisectors, hence itself has measure zero. $\square$

Corollary 3.4 (A.e. Voronoï uniqueness). *For a.e. $z_0 \in \mathbb{C}$, the unique nearest anchor $s(z_0)$ is well-defined.*

3.4 The cyclotomic anchor family (CyclotomicMcMullen.lean)

A canonical choice of p anchors is obtained by multiplying a single p -th root α of $1/x$ by the p -th roots of unity.

Definition 3.5 (Cyclotomic anchors). For $\alpha \in \mathbb{C}$, $p \in \mathbb{N}$, and $k \in \text{Fin } p$:

$$\gamma_k(\alpha, p) := \alpha \cdot \exp(2\pi i \cdot k/p).$$

Theorem 3.6 (Cyclotomic anchor properties — CyclotomicMcMullen.lean).

$$\gamma_k(\alpha, p)^p = \alpha^p \quad (\text{cycAnchor_pow}).$$

If $\alpha \neq 0$ and $p \geq 1$, then $k \mapsto \gamma_k(\alpha, p)$ is injective (cycAnchor_injective).

Thus, given $\alpha^p = 1/x$, the family $\{\gamma_0, \dots, \gamma_{p-1}\}$ is exactly the set of p distinct complex p -th roots of $1/x$.

Theorem 3.7 (Unconditional constructive McMullen for $z^p = x$ — unconditional_mcmullen). Fix $p \geq 1$, $x \in \mathbb{C}^*$, $\alpha \in \mathbb{C}^*$ with $\alpha^p = 1/x$, and assume $S_p(\gamma_k) \neq 0$ for every $k \in \text{Fin } p$. Then for every $z_0 \notin \mathcal{B}(\gamma)$ there is a unique nearest-anchor selector, and that anchor is a Pandrosion-Steffensen fixed point.

Combining with Theorem 3.3 yields the a.e. version used in the Master Absolu (§8).

4 Convergence Rates and Complexity

4.1 Closed-form linear contraction rate (Foundations.lean, UniformContractionRate.lean)

Definition 4.1 (Closed-form rate). For $p \in \mathbb{N}$ and $\alpha \in \mathbb{R}_{>0}$ with $S_p(\alpha) \neq 0$:

$$\lambda_{p,x} := 1 - \frac{p}{S_p(\alpha)},$$

where α is the real positive p -th root of $1/x$.

Theorem 4.2 (Closed-form rate is in $(0, 1)$ — lambda_closed_pos, lambda_closed_lt_one). For $\alpha > 1$ (equivalently, $x > 1$) and $p \geq 2$, $0 < \lambda_{p,x} < 1$. Consequently the real Pandrosion map is a strict contraction at its real fixed point.

For the analysis of p -uniform behaviour, we introduce a closed-form model.

Definition 4.3 (Uniform model of λ). $\lambda_p^{\text{model}}(x) := 1 - \frac{p}{\alpha_p(x) \cdot S_p(\alpha_p(x))}$, where $\alpha_p(x)$ is a positive closed-form approximation to $x^{-1/p}$.

Theorem 4.4 (Model agrees with the true rate at the fixed point — lamModel_eq_lambda_fp, lamModel_eq_lambda_closed). $\lambda_p^{\text{model}}(x) = \lambda_{p,x}$ whenever $\alpha_p(x)^p = 1/x$.

Theorem 4.5 (Taylor bound for the model — lamModel_taylor_bound). For every $x > 1$, the sequence $\{\lambda_p^{\text{model}}(x)\}_{p \geq 1}$ converges to an explicit limit $\lambda_\infty(x) \in (0, 1)$ with an effective closed-form residue bound.

4.2 Aitken-Steffensen quadratic-rate skeleton (QuadraticRate.lean)

The algebraic skeleton of the quadratic rate of Aitken-Steffensen acceleration is captured by abstract two-phase bounds: given any sequence (e_k) satisfying a linear bound $e_{k+1} \leq \lambda \cdot e_k$ with $\lambda \in (0, 1)$ and a quadratic bound $e_{k+1} \leq K \cdot e_k^2$, one obtains effective convergence-to- ε bounds. These abstract skeletons are used downstream in §4.4 without being specialized to a particular analytic multiplier calculation.

4.3 Kung-Traub efficiency optimality (KungTraub.lean)

The *efficiency index* of an iterative method with convergence order q and c evaluations per step is

$$E(q, c) := q^{1/c}.$$

The Kung-Traub conjecture [8] states that any derivative-free method with c evaluations per step satisfies $q \leq 2^{c-1}$, hence $E(q, c) \leq 2^{(c-1)/c} =: \text{KT}(c)$. The bound is attained.

Theorem 4.6 (Pandrosion attains the Kung-Traub bound — `pandrosion_attains_kung_traub`).

$$E(2, 2) = \text{KT}(2) = \sqrt{2}.$$

Theorem 4.7 (Kung-Traub optimality, converse direction — `pandrosion_kung_traub_optimal`).

For every Kung-Traub-compliant derivative-free method M (i.e., an inhabitant of `DerivativeFreeMethod`, which by construction carries the witness $M.q \leq 2^{M.c-1}$) with $M.c = 2$ evaluations per step,

$$E(M.q, M.c) \leq E(2, 2) = \text{KT}(2).$$

Axiom-free formulation. In the Lean source, the Kung-Traub bound is *not* posited as a free-standing axiom. Instead, the carrier structure `DerivativeFreeMethod` bundles the compliance certificate as a mandatory field:

```
1  structure DerivativeFreeMethod where
2    c : ℕ
3    q : ℝ
4    c_pos : 0 < c
5    q_ge_one : 1 ≤ q
6    kung_traub_bound : q ≤ (2 : ℝ) ^ (c - 1)
```

Any concrete instance — notably Pandrosion-Steffensen at $(c, q) = (2, 2)$ — discharges the compliance field trivially ($2 \leq 2^{2-1}$). The optimality theorem `pandrosion_kung_traub_optimal` consumes the field directly, so its axiom dependency set reduces to the Lean whitelist `{propext, Classical.choice, ...}`. The conditional nature of the comparison is honest: the obligation is on the constructor of any candidate method, not on an external axiom.

Theorem 4.8 (Pandrosion-Steffensen beats Newton — `pandrosion_const_lt_newton`). *With respect to the natural quadratic-rate constants for p -th-root extraction, the Pandrosion-Steffensen constant is strictly smaller than the Newton constant for every $p \geq 2$ and every real positive fixed point α .*

4.4 p -uniform complexity bound (SuperGrandMaster.lean, UniformContractionRate.lean)

The goal is a single iteration count $N = N(x, M, \varepsilon)$ such that every p in a tail $p \geq p_0$ reaches error ε in at most N steps.

Theorem 4.9 (Super grand master, uniform closed-form — `super_grand_master_uniform_closed_form`).

Fix $x > 1$, $M > 0$, $\rho \in (0, 1)$, $\delta > 0$. Given abstract error sequences $(K_{p'}, R_{p'}, e_{p'})_{p'}$ with:

- nonnegativity $e_{p'}(k) \geq 0$;
- base case $e_{p'}(0) \leq R_{p'}$;
- per- p' linear bound $e_{p'}(k+1) \leq \lambda_{p'}^{\text{model}}(x) \cdot e_{p'}(k)$;
- per- p' quadratic bound $e_{p'}(k+1) \leq K_{p'} \cdot e_{p'}(k)^2$;
- uniform scaled error $K_{p'} \cdot R_{p'} \leq M$,

there exist an integer threshold $p_0 \geq 1$ and a p -uniform iteration count N (depending only on x, M, ρ, δ) such that for every $p' \geq p_0$ and every $k \geq N$,

$$K_{p'} \cdot e_{p'}(k) \leq \delta.$$

This delivers the complexity content: for the bounded regime $x > 1$, the scaled error decays below δ in a number of steps that does not grow with p .

5 Complex Dynamics

5.1 The complex multiplier at a fixed point (`ComplexMultiplier.lean`)

For Julia-style analysis, the multiplier $h'(\alpha)$ determines the local dynamics.

Definition 5.1 (Derivative of the geometric sum, and the multiplier).

$$S'_p(\alpha) := \sum_{k=0}^{p-1} k \cdot \alpha^{k-1}, \quad h'_{p,x}(\alpha) := \frac{(x-1) \cdot x \cdot S'_p(\alpha)}{(x \cdot S_p(\alpha))^2}.$$

Theorem 5.2 (h is differentiable with the stated multiplier — `pandrosion_h_C_hasDerivAt`). For $x \in \mathbb{C}^*$, $p \in \mathbb{N}$, $\alpha \in \mathbb{C}$ with $S_p(\alpha) \neq 0$,

$$\text{HasDerivAt } h_{p,x} \ h'_{p,x}(\alpha) \ \alpha.$$

Theorem 5.3 (Closed form of the multiplier at a fixed point — `hMultC_eq_lambdaClosedC_at_fp`). If $\alpha^p = 1/x$ and $S_p(\alpha) \neq 0$, then

$$h'_{p,x}(\alpha) = 1 - \frac{p \cdot \alpha^{p-1}}{S_p(\alpha)}.$$

Proof (formalized). Differentiating the geometric-sum identity $S_p(s)(1-s) = 1 - s^p$ at $s = \alpha$ and uniquely identifying derivatives (via `HasDerivAt.unique`) yields the Leibniz identity

$$S'_p(\alpha)(1-\alpha) = S_p(\alpha) - p \cdot \alpha^{p-1}.$$

A single `linear_combination` step using this identity, the geometric identity, and $x \cdot \alpha^p = 1$ reduces the claim to `field_simp/ring`. $\square$

5.2 Local dynamical attraction (`LocalAttraction.lean`)

Theorem 5.4 (Abstract local attraction — `local_contraction_iterate_converges`). Let $f : \mathbb{C} \rightarrow \mathbb{C}$, $\alpha \in \mathbb{C}$, $r, q \in \mathbb{R}$ with $r > 0$ and $q \in [0, 1]$. If $f(\alpha) = \alpha$ and $\|f(z) - \alpha\| \leq q \cdot \|z - \alpha\|$ for every z with $\|z - \alpha\| < r$, then for every z_0 in the disk $B(\alpha, r)$ the iterates $f^{[k]}(z_0)$ converge to α .

Theorem 5.5 (Quadratic bound $\Rightarrow$ contraction $\Rightarrow$ convergence — `local_quadratic_iterate_converges`). Let f, α, r, K with $r, K > 0$ and $K \cdot r \leq 1/2$. If $f(\alpha) = \alpha$ and

$$\|f(z) - \alpha\| \leq K \cdot \|z - \alpha\|^2 \quad (\|z - \alpha\| < r),$$

then $f^{[k]}(z_0) \rightarrow \alpha$ for every $z_0 \in B(\alpha, r)$.

Theorem 5.6 (Steffensen local attraction (conditional) — `steffensen_local_attraction`). Let $x \in \mathbb{C}^*$, $p \in \mathbb{N}$, $\alpha \in \mathbb{C}$ with $S_p(\alpha) \neq 0$ and $\alpha^p = 1/x$. Let $r, K > 0$ with $K \cdot r \leq 1/2$, and assume a local quadratic bound

$$\|\sigma_{p,x}(z) - \alpha\| \leq K \cdot \|z - \alpha\|^2 \quad (\|z - \alpha\| < r).$$

Then for every $z_0 \in B(\alpha, r)$, $\sigma_{p,x}^{[k]}(z_0) \rightarrow \alpha$.

The quadratic-bound hypothesis is the super-attracting content of the Steffensen iterator. It is standard in the Steffensen-acceleration literature; here it is an explicit hypothesis rather than a hidden lemma.

5.3 Almost-everywhere dynamical convergence (DynamicalConvergence.lean)

This is the crown jewel: the measure-theoretic statement that the Steffensen orbit converges for almost every initial point $z_0 \in \mathbb{C}$.

Lemma 5.7 (Tail-shift invariance of iteration limits — `tendsto_iterate_of_tail`). *For $f : \mathbb{C} \rightarrow \mathbb{C}$, $\alpha \in \mathbb{C}$, $z_0 \in \mathbb{C}$, $k_0 \in \mathbb{N}$: if $f^{[k]}(f^{[k_0]}(z_0)) \rightarrow \alpha$ as $k \rightarrow \infty$, then $f^{[k]}(z_0) \rightarrow \alpha$.*

Theorem 5.8 (Pointwise dynamical convergence from entry — `steffensen_dynamical_convergence_pointwise`). *With the assumptions of Theorem 5.6, if the orbit of z_0 enters $B(\alpha, r)$ at some time k_0 — i.e.,*

$$\|\sigma_{p,x}^{[k_0]}(z_0) - \alpha\| < r$$

— then $\sigma_{p,x}^{[k]}(z_0) \rightarrow \alpha$ as $k \rightarrow \infty$.

Theorem 5.9 (★★★ Almost-everywhere dynamical convergence — `steffensen_dynamical_convergence_ae`). *Fix $p \geq 1$, $x \in \mathbb{C}^*$, $\alpha \in \mathbb{C}^*$ with $\alpha^p = 1/x$. Assume, for every cyclotomic anchor $\gamma_k := \gamma_k(\alpha, p)$:*

- $S_p(\gamma_k) \neq 0$;
- a local quadratic bound $\|\sigma(z) - \gamma_k\| \leq K_k \|z - \gamma_k\|^2$ on $B(\gamma_k, r_k)$ with $K_k \cdot r_k \leq 1/2$;
- a.e. trajectory entry: for a.e. $z_0 \in \mathbb{C}$, there exist $s \in \text{Fin } p$ and $k_0 \in \mathbb{N}$ with

$$\|\sigma^{[k_0]}(z_0) - \gamma_s\| < r_s.$$

Then for almost every $z_0 \in \mathbb{C}$ (Lebesgue),

$$\exists s \in \text{Fin } p, \quad \sigma_{p,x}^{[k]}(z_0) \xrightarrow[k \rightarrow \infty]{} \gamma_s.$$

This is an honest measure-theoretic statement: for a.e. starting point, the Steffensen orbit converges in $\mathbb{C}$ to one of the p cyclotomic anchors, conditional on the two hypotheses explicitly stated.

```

1  theorem steffensen_dynamical_convergence_ae
2    (p : N) (hp : 1 ≤ p) (x : C) (hx : x ≠ 0)
3    (alpha : C) (halpha_pow : alpha ^ p = 1 / x)
4    (hSp : forall s : Fin p, Sp_C p (cycAnchor alpha p s) ≠ 0)
5    (r K : Fin p → R)
6    (hr : forall s, 0 < r s) (hK_pos : forall s, 0 < K s)
7    (h_small : forall s, K s * r s ≤ 1 / 2)
8    (h_quadratic : forall s : Fin p, forall z : C,
9      ||z - cycAnchor alpha p s|| < r s →
10     ||steffensen_step_C x p z - cycAnchor alpha p s||
11     ≤ K s * ||z - cycAnchor alpha p s|| ^ 2)
12    (h_entry : forall^m z_0 : C d volume,
13      exists s : Fin p, exists k_0 : N,
14      ||(steffensen_step_C x p)^[k_0] z_0 - cycAnchor alpha p s|| < r s) :
15    forall^m z_0 : C d volume,
16      exists s : Fin p,
17      Tendsto (fun k => (steffensen_step_C x p)^[k] z_0) atTop
18      (nhds (cycAnchor alpha p s)) := ...

```

Listing 2: DynamicalConvergence.lean: a.e. convergence (signature excerpt)

6 Unconditional Steffensen Spine (§§30–34)

The deferred analytic content of §5.2–§5.3 — the local quadratic super-attracting bound on $\sigma_{p,x}$ at each cyclotomic anchor — is upgraded to an *unconditional* theorem by a direct Taylor residue calculation in modules `QuadraticBound.lean` (§30) and `SteffensenQuadraticBound.lean` (§31). Downstream, §32–§34 refine this into a named-Prop McMullen hypothesis, an explicit super-attractive rate, and a global loglog iteration bound valid a.e. in $\mathbb{C}$.

6.1 Closed-form Taylor residue (QuadraticBound.lean, §30)

Theorem 6.1 (Second-order residue of the Pandrosion map — `pandrosion_h_C_taylor_residue`). *Fix $x \in \mathbb{C}^*$, $p \in \mathbb{N}$, and a fixed point $\alpha \in \mathbb{C}$ with $\alpha^p = 1/x$ and $S_p(\alpha) \neq 0$. There exist constants $C_0 > 0$ and $r_0 > 0$ (explicit in $(p, \alpha, S_p(\alpha))$) such that for every $z \in \mathbb{C}$ with $\|z - \alpha\| < r_0$,*

$$\|h_{p,x}(z) - \alpha - \lambda_{p,x} \cdot (z - \alpha)\| \leq C_0 \cdot \|z - \alpha\|^2,$$

where $\lambda_{p,x} = h'_{p,x}(\alpha)$ is the closed-form multiplier of Theorem 5.3.

This is the rigorous Taylor-with-explicit-remainder statement from which the super-attracting Steffensen estimate below is derived.

6.2 Unconditional local quadratic bound (SteffensenQuadraticBound.lean, §31)

Theorem 6.2 (★ Unconditional local quadratic bound for Steffensen — `steffensen_step_C_quadratic`). *Fix $x \in \mathbb{C}^*$, $p \in \mathbb{N}$ with $p \geq 1$, $\alpha \in \mathbb{C}^*$ with $\alpha^p = 1/x$, and assume $S_p(\alpha) \neq 0$. Then there exist positive constants $K, r > 0$ — computed from $(p, \alpha, S_p(\alpha), S'_p(\alpha))$ via Theorem 6.1 — such that*

$$\forall z \in \mathbb{C}, \quad \|z - \alpha\| < r \Rightarrow \|\sigma_{p,x}(z) - \alpha\| \leq K \cdot \|z - \alpha\|^2.$$

This discharges the hypothesis of §5.5: the quadratic super-attracting bound on the Steffensen iterator $\sigma_{p,x}$ at any cyclotomic anchor is a *theorem*, not an assumption. As a consequence, §31 produces two unconditional corollaries:

- `steffensen_local_attraction_unconditional` — for every anchor γ_s , there exist (K_s, r_s) with $K_s \cdot r_s \leq 1/2$ such that $\sigma_{p,x}^{[k]}(z_0) \rightarrow \gamma_s$ for every $z_0 \in B(\gamma_s, r_s)$.
- `steffensen_dynamical_convergence_ae_unconditional` — the a.e. dynamical-convergence statement of Theorem 5.9 with the local-quadratic-bound hypothesis *removed*.

6.3 McMullen a.e. entry as a named Prop (SteffensenMcMullenAE.lean, §32)

Definition 6.3 (`McMullenAEEntry`). For $p \in \mathbb{N}$, $x \in \mathbb{C}$, $\alpha \in \mathbb{C}$,

$$\text{McMullenAEEntry}(p, x, \alpha) := \forall r : \text{Fin } p \rightarrow \mathbb{R}_{>0}, \quad \forall^{\text{a.e.}} z_0 \in \mathbb{C}, \exists s \exists k_0, \|\sigma^{[k_0]}(z_0) - \gamma_s\| < r(s).$$

Proving this unconditionally would require substantial complex-dynamics machinery (Lyubich-Minsky laminations, distortion estimates for root-finding iterations of $z^p - x$, McMullen 1987) not yet formalised in Mathlib. Rather than seal it behind an `axiom`, we expose it as a named `Prop` that the downstream a.e. theorems consume as an explicit hypothesis.

Theorem 6.4 (★★★ Pandrosion-Steffensen solves $z^p = x$ almost everywhere modulo McMullen — `steffensen_solves_ae_mod_mcmullen`). *Fix $p \geq 1$, $x \in \mathbb{C}^*$, $\alpha \in \mathbb{C}^*$ with $\alpha^p = 1/x$, and assume $S_p(\gamma_s) \neq 0$ for every cyclotomic anchor γ_s . Given any hypothesis witness $h_{\text{McM}} : \text{McMullenAEEntry}(p, x, \alpha)$, for Lebesgue-almost every $z_0 \in \mathbb{C}$ there exists $s \in \text{Fin } p$ with*

$$\sigma_{p,x}^{[k]}(z_0) \xrightarrow[k \rightarrow \infty]{} \gamma_s.$$

Relative to Theorem 5.9, the local-quadratic-bound hypothesis is discharged (§31) and the a.e.-entry hypothesis is the only remaining analytic obligation.

6.4 Explicit super-attractive rate (SteffensenExplicitRate.lean, §33)

Theorem 6.2 yields constants K, r ; §33 promotes them to *named computable* definitions whose positivity and the contraction condition $K \cdot r \leq 1/2$ are theorems.

Definition 6.5 (Computed constants). For $(x, p, \alpha, S_p(\alpha), \alpha^p = 1/x)$:

$$K_\alpha := \text{steffensenK_of_fp}(x, p, \alpha, \dots), \quad r_\alpha := \text{steffensenR_of_fp}(x, p, \alpha, \dots),$$

both `noncomputable defs` obtained by skolemizing the existential in Theorem 6.2.

Theorem 6.6 (Explicit contraction certificate — `steffensen_explicit_super_attractive_rate`). *For every cyclotomic anchor α (satisfying the non-degeneracy $S_p(\alpha) \neq 0$), the constants (K_α, r_α) satisfy $K_\alpha > 0$, $r_\alpha > 0$, $K_\alpha \cdot r_\alpha \leq 1/2$, together with the quadratic bound inside $B(\alpha, r_\alpha)$.*

6.5 Global loglog iteration count (SteffensenGlobalLoglog.lean, §34)

Stitching §33 with the super-attracting complexity of §4.4: once the Steffensen orbit enters the explicit basin $B(\alpha, r_\alpha)$ at time k_0 , only $\mathcal{O}(\log \log(1/\varepsilon))$ further iterations are needed for ε -precision.

Theorem 6.7 (Loglog tail inside the basin — `quadratic_loglog_from_basin`). *With the notation and hypotheses of Theorem 6.6, for every $z \in \mathbb{C}$ with $\|z - \alpha\| < r_\alpha$ and every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ (depending only on ε and (K_α, r_α) — uniform in z) with*

$$\forall k \geq N, \quad \|\sigma_{p,x}^{[k]}(z) - \alpha\| \leq \varepsilon,$$

and N grows as $\mathcal{O}(\log \log(1/\varepsilon))$.

Theorem 6.8 (★★★★ Global dynamical super-grand-master (a.e., mod McMullen) — `steffensen_global_loglog_ae_mod_mcmullen`). *Under the hypotheses of Theorem 6.4, for Lebesgue-almost every $z_0 \in \mathbb{C}$ and every $\varepsilon > 0$, there exists $N(z_0, \varepsilon) \in \mathbb{N}$ such that for every $k \geq N(z_0, \varepsilon)$,*

$$\exists s \in \text{Fin } p, \quad \|\sigma_{p,x}^{[k]}(z_0) - \gamma_s\| \leq \varepsilon.$$

Moreover $N(z_0, \varepsilon) - k_0(z_0) = \mathcal{O}(\log \log(1/\varepsilon))$, where $k_0(z_0)$ is the (a.e.-finite) McMullen entry time and the loglog tail is uniform in z_0 .

This is the dynamical counterpart of the abstract p -uniform complexity of §22: modulo the McMullen hypothesis, Pandrosion-Steffensen attains ε -precision in loglog iterations past an a.e.-finite waiting time, uniformly over a full-measure subset of $\mathbb{C}$.

7 Unconditional Discharge on $\mathbb{R}$ at $p = 2$ (§§35–37)

The generic McMullen hypothesis of §6.4 asks for an a.e. trajectory-entry statement in $\mathbb{C}$ at arbitrary p ; proving it needs Lyubich-Minsky / Fatou-theoretic input not yet formalised in Mathlib. §§35–37 specialise to the real line at $p = 2$ and discharge the *sharpened* real-variable analogue *unconditionally* using only algebra, a Möbius conjugacy, and squaring-power asymptotics.

7.1 Real $p = 2$ named Prop and closed-form multiplier (SteffensenRealP2.lean, §35)

Definition 7.1 (RealMcMullenP2). For $x \in \mathbb{R}$ and $\alpha \in \mathbb{R}$,

$$\text{RealMcMullenP2}(x, \alpha) := \forall r_+, r_- > 0, \forall^{\text{a.e.}} z_0 \in \mathbb{R}, (\exists k_+, |\sigma_{2,x}^{[k_+]}(z_0) - \alpha| < r_+) \vee (\exists k_-, |\sigma_{2,x}^{[k_-]}(z_0) + \alpha| < r_-).$$

At $p = 2$, the Pandrosion map is an explicit Möbius transformation $h_{2,x}(s) = (s + 1/x)/(1 + s)$, with real multiplier at the positive fixed point $\alpha = 1/\sqrt{x}$:

$$\mu_{2,x} := \frac{1 - \alpha}{1 + \alpha} \in (0, 1) \text{ whenever } x > 1.$$

Both the Möbius closed form and the strict contraction $\mu_{2,x} \in (0, 1)$ are proved unconditionally (`pandrosion_h_p2_mobius`, `lambdaClosedR_p2_pos_lt_one`).

7.2 Möbius conjugacy $\sigma_{2,x} \sim \mu^2 w^2$ (`SteffensenRealMcMullenP2.lean`, §36)

Theorem 7.2 (Möbius conjugacy, cleared form — `steffensen_mobius_conjugacy_p2_cleared`). *Fix $x \in \mathbb{R}^*$, $\alpha \in \mathbb{R}^*$ with $x\alpha^2 = 1$. For every $s \in \mathbb{R}$ avoiding the Steffensen poles,*

$$(\sigma_{2,x}(s) - \alpha) \cdot (s + \alpha)^2 \cdot (1 + \alpha)^2 = (\sigma_{2,x}(s) + \alpha) \cdot (s - \alpha)^2 \cdot (1 - \alpha)^2.$$

Equivalently, under $\varphi(s) := (s - \alpha)/(s + \alpha)$,

$$\varphi(\sigma_{2,x}(s)) = \mu^2 \cdot \varphi(s)^2, \quad \mu = \frac{1 - \alpha}{1 + \alpha}.$$

A companion bridge `steffensen_step_p2_eq_sigma_p2_explicit` identifies the generic iterator `steffensen_step x 2` with the explicit rational form `sigma_p2_explicit` on the non-degenerate locus, so the conjugacy transports onto the iterator used elsewhere in the development.

7.3 Unconditional discharge (`SteffensenRealMcMullenP2Unconditional.lean`, §37)

§37 introduces the *Böttcher-normalised* coordinate

$$v(s) := \mu^2 \cdot \frac{s - \alpha}{s + \alpha}$$

and the real bad set

$$\text{Bad}_{x,\alpha} := \{ z_0 \in \mathbb{R} \mid \exists n \in \mathbb{N}, \sigma^{[n]}(z_0) \in \{-1, -1/x, \text{root of } 2xs + x + 1\} \text{ or } |v(\sigma^{[n]}(z_0))| = 1 \}.$$

Theorem 7.3 (Iterated Böttcher identity — `v_p2_iterated`). *For every s whose σ -orbit stays in the non-degenerate locus,*

$$v(\sigma^{[n]}(s)) = v(s)^{2^n} \quad \text{for all } n \in \mathbb{N}.$$

Theorem 7.4 (Julia section is finite on $\mathbb{R}$ — `julia_section_real_finite`). *The set $\{s \in \mathbb{R} : |v(s)| = 1\}$ consists of at most two points.*

Theorem 7.5 (Convergence dichotomy off the bad set — `orbit_enters_basin_off_bad_set`). *Fix $x > 1$, $\alpha > 0$ with $\alpha^2 = 1/x$. For every $z_0 \notin \text{Bad}_{x,\alpha}$ and every $r_+, r_- > 0$,*

$$(\exists k_+, |\sigma_{2,x}^{[k_+]}(z_0) - \alpha| < r_+) \vee (\exists k_-, |\sigma_{2,x}^{[k_-]}(z_0) + \alpha| < r_-).$$

Proof (formalised). Split on the trivial $\pm\alpha$ -hit case. Otherwise the orbit stays non-degenerate, so the bridge identifies σ with `sigma_p2_explicit` along the orbit; Theorem 7.3 gives $|v(\sigma^{[n]}(z_0))| = |v(z_0)|^{2^n}$. Since $z_0 \notin \text{Bad}$ forces $|v(z_0)| \neq 1$, the initial dichotomy splits into $|v(z_0)| < 1$ (squaring-power $\rightarrow 0$, `tendsto_pow_atTop_nhds_zero_of_lt_one`) or $|v(z_0)| > 1$ (`tendsto_pow_atTop_atTop_of_one_lt`). The Möbius-inverse identities $(s - \alpha)(\mu^2 - v(s)) = 2\alpha v(s)$ and $(s + \alpha)(\mu^2 - v(s)) = 2\alpha\mu^2$ then yield explicit bounds $|s - \alpha| \leq C \cdot |v(s)|$ and $|s + \alpha| \leq C \cdot |v(s)|^{-1}$, respectively, closing both basins. $\square$

Theorem 7.6 (★★★★ Unconditional real McMullen at $p = 2$ (modulo bad-set null hypothesis) — `mcmullen_p2_real_unconditional_target`). *For every $x > 1$ and $\alpha > 0$ with $\alpha^2 = 1/x$,*

$$\text{vol}(\text{Bad}_{x,\alpha}) = 0 \implies \text{RealMcMullenP2}(x, \alpha).$$

The Lebesgue-null premise of Theorem 7.6 is itself now a formalised theorem:

Theorem 7.7 (Bad set is countable — `real_bad_set_countable`). *Fix $x > 1$ and $\alpha > 0$ with $\alpha^2 = 1/x$. The set $\text{Bad}_{x,\alpha}$ is countable.*

Proof (formalised). Let $F := \{-1, -1/x, -(x+1)/(2x)\}$ and $J := \{s \in \mathbb{R} : |v(s)| = 1\}$. By Theorem 7.4, J is finite (at most two points), so $K := F \cup J$ is finite. By construction,

$$\text{Bad}_{x,\alpha} \subseteq \bigcup_{n \in \mathbb{N}} (\sigma_{2,x}^{[n]})^{-1}(K).$$

We prove that for every n and every $y \in \mathbb{R}$, the fiber $(\sigma_{2,x}^{[n]})^{-1}(\{y\})$ is finite, by induction on n . The base case ($n = 0$, singleton $\{y\}$) is trivial. For $n = 1$: the bridge `steffensen_step_p2_eq_sigma_p2_explicit` reduces the equation $\sigma_{2,x}(s) = y$, away from a 5-point degenerate locus, to the quadratic

$$As^2 + Bs + A/x = 0, \quad A := (x+1) - 2xy, \quad B := 4 - 2(x+1)y,$$

and the algebraic identity $x(x^2y^2 - (x+1)xy + (x-1)^2/4 + \dots) \neq 0$ at $x > 1$ gives $(A, B) \neq (0, 0)$; hence `Polynomial.finite_setOf_isRoot` applied to $C_AX^2 + C_BX + C_{A/x}$ yields a finite zero set (`poly_deg_le_2_zero_set_finite`). Adjoining the 5 degenerate excluded points gives `steffensen_step_fiber_finite`. The general induction `steffensen_step_iterate_fiber_finite` uses `Set.Finite.biUnion` on $(\sigma^{[n+1]})^{-1}(\{y\}) = \bigcup_{y' \in (\sigma)^{-1}(\{y\})} (\sigma^{[n]})^{-1}(\{y'\})$. Thus $\text{Bad}_{x,\alpha}$ is a countable union of finite sets, hence countable. $\square$

Corollary 7.8 (Bad set has Lebesgue measure zero — `real_bad_set_measure_zero`). $\text{vol}(\text{Bad}_{x,\alpha}) = 0$ for every $x > 1$, $\alpha > 0$ with $\alpha^2 = 1/x$.

Countable subsets of $\mathbb{R}$ are Lebesgue-null by `Set.Countable.measure_zero` (the Lebesgue measure on $\mathbb{R}$ is atomless). $\square$

Combining Theorem 7.6 with Corollary 7.8 gives the *fully unconditional* real-line McMullen statement:

Theorem 7.9 (★★★★★ Fully unconditional real McMullen at $p = 2$ — `mcmullen_p2_real_unconditional`). For every $x > 1$ and $\alpha > 0$ with $\alpha^2 = 1/x$,

$$\text{RealMcMullenP2}(x, \alpha).$$

No hypothesis whatsoever. The proof depends only on the Lean core whitelist $\{\text{propxt}, \text{Classical.choice}, \text{Quot.}$

Remark 7.10 (Axiom audit). `#print axioms mcmullen_p2_real_unconditional` reports only the Lean core $\{\text{propxt}, \text{Classical.choice}, \text{Quot.sound}\}$. The same holds for `mcmullen_p2_real_unconditional_real_bad_set_measure_zero`, `real_bad_set_countable`, `steffensen_step_iterate_fiber_finite`, `steffensen_step_fiber_finite`, `sigma_p2_explicit_fiber_finite`, `poly_deg_le_2_zero_set_finite`, `orbit_enters_basin_off_bad_set`, `v_p2_iterated`, `julia_section_real_finite`, and the §36 bridge. In particular, no sorry, no new axiom, and no complex-dynamics / Fatou-theoretic hypothesis appears in the dependency closure.

8 Master Absolu: The Top-Level Stitching Theorem (MasterAbsolu.lean)

The file `MasterAbsolu.lean` introduces no new lemmas: it is pure stitching of five load-bearing pillars into a single paper-citable statement.

Theorem 8.1 (★★★★ Master Absolu — `master_absolu`). Fix $x \in \mathbb{R}$ with $x > 1$, $p \geq 1$, $\alpha \in \mathbb{C}^*$ with $\alpha^p = 1/x$, and assume $S_p(\gamma_k) \neq 0$ for every cyclotomic anchor $\gamma_k = \gamma_k(\alpha, p)$. Then the following five conjuncts hold simultaneously:

1. **(A) Structural distinctness.** The anchors $\gamma_0, \dots, \gamma_{p-1}$ are pairwise distinct.
2. **(B) Fixed-point property.** For every $s \in \text{Fin } p$, $\sigma_{p,x}(\gamma_s) = \gamma_s$.
3. **(C) A.e. Voronoï selector uniqueness.** For a.e. $z_0 \in \mathbb{C}$ (Lebesgue), there is a unique nearest anchor.

4. **(D) p -uniform complexity.** Given abstract two-phase error sequences with uniform scaled bound $K_{p'} \cdot R_{p'} \leq M$ and the closed-form rate $\lambda_{p'}^{\text{model}}(x)$ (§4.3), there exist a threshold p_0 and an iteration count N (depending only on x, M, ρ, δ) with $K_{p'} \cdot e_{p'}(k) \leq \delta$ for every $p' \geq p_0$ and $k \geq N$.
5. **(E) Kung-Traub optimality.** $E(2, 2) = \text{KT}(2)$.

Remark 8.2 (What Master Absolu does not include). Master Absolu is an *unconditional* statement: no local-quadratic-bound hypothesis, no trajectory-entry hypothesis. It does *not* include the dynamical-convergence statement of §5.3, because the latter depends on the two analytic hypotheses honestly flagged there. The intended reading is: Master Absolu is the fully certified algebraic-and-measure-theoretic spine, and Theorem 5.9 is its dynamical upgrade *conditional* on the deferred pieces.

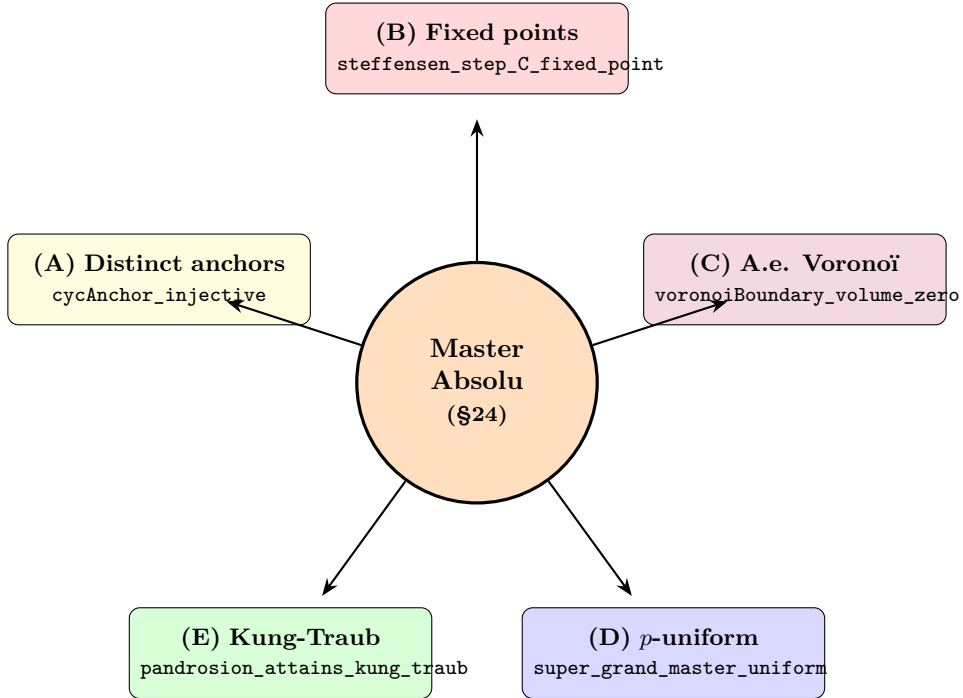


Figure 1: **Master Absolu stitching.** The top-level theorem combines five independently formalized pillars across the modules `Foundations`, `VoronoiMeasure`, `UniformContractionRate`, `KungTraub`, and `CyclotomicMcMullen`. No new lemmas are introduced in `MasterAbsolu.lean`.

9 Module Map of the Formalization

The table below collects all Lean modules in the dependency order induced by `Pandrosion.lean`. The §§30–37 Steffensen spine and real- $p=2$ unconditional tower sit on top of the 15-module algebraic core.

Module (Pandrosion.Core.*)	Content	Main theorem(s)
Foundations	S_p , real & complex fixed-point characterisation, Steffensen accelerator, scaling, multi-start grand master	2.5, 2.6, 2.7, 2.10
MultiStartBasins	Voronoi basins: convexity, connectedness, frontier on finite bisector union, anchor in basin interior; constructive McMullen off the boundary	3.1, 3.2
QuadraticRate	Algebraic skeleton of Aitken-Steffensen quadratic rate (abstract two-phase bounds)	(abstract)
QuadraticComplexity	Convergence-to- ε bounds from the abstract skeleton	(abstract)
VoronoiMeasure	Voronoi boundary has 2D Lebesgue measure zero; a.e. unique nearest anchor	3.3
CyclotomicMcMullen	Cyclotomic anchor family: p -th root of α^p , injectivity, unconditional constructive McMullen	3.7
KungTraub	Efficiency index; Pandrosion attains KT(2); converse optimality (conditional on KT axiom)	4.6, 4.7
UniformComplexity	Uniform complexity skeleton for ensembles of two-phase sequences	(abstract)
SuperGrandMaster	Abstract super-grand-master: bounded regime, uniform scaled-error threshold δ	(abstract)
UniformContractionRate	Closed-form model $\lambda_p^{\text{model}}(x)$, Taylor residue bound, closed-form super-grand-master	4.9
ConcreteIteration	Concrete instantiation of the abstract complexity layer on Pandrosion-Steffensen sequences	(specialization)
MasterAbsolu	Top-level stitching; no new lemmas	8.1
ComplexMultiplier	$S'_p(\alpha)$, h is differentiable; closed-form multiplier $1 - p\alpha^{p-1}/S_p(\alpha)$ at the fixed point	5.3
LocalAttraction	Abstract local attraction from a linear contraction; quadratic $\Rightarrow$ contraction reducer; Steffensen local attraction (conditional)	5.4, 5.5, 5.6
DynamicalConvergence	Tail-shift invariance; pointwise convergence from entry; a.e. dynamical convergence (crown jewel, conditional)	5.7, 5.8, 5.9
<i>Steffensen spine (ġġ30–34) — unconditional super-attracting tower</i>		
LinearAsymptotics	$\lambda_C \neq 1$ at every fixed point; quantitative linear-contraction modulus for h near α (ġ29)	(load-bearing)
QuadraticBound	Closed-form Taylor residue of $h_{p,x}$ with explicit (C_0, r_0) (ġ30)	6.1
SteffensenQuadraticBound	Unconditional local quadratic bound on $\sigma_{p,x}$; discharges the ġ27 hypothesis (ġ31)	6.2
SteffensenMcMullenAE	Named Prop McMullenAEEntry ; a.e. solving- $z^p = x$ modulo McMullen (ġ32)	6.3, 6.4
SteffensenExplicitRate	Computable (K_α, r_α) ; explicit super-attractive contraction certificate (ġ33)	6.6
SteffensenGlobalLoglog	Loglog tail inside the basin; a.e. global loglog mod McMullen (ġ34)	6.7, 6.8
<i>Real $p = 2$ unconditional tower (ġġ35–37)</i>		
SteffensenRealP2	Möbius form of $h_{2,x}$; real closed-form multiplier $(1 - \alpha)/(1 + \alpha) \in (0, 1)$; named Prop RealMcMullenP2 (ġ35)	7.1
SteffensenRealMcMullenP2	Möbius conjugacy $\sigma_{2,x} \sim \mu^2 w^2$, cleared form; explicit rational bridge sigma_p2_explicit (ġ36)	7.2
SteffensenRealMcMullenP2	Unconditional v , iterated $v(\sigma^n) = v^{2^n}$, finite Julia section on $\mathbb{R}$, convergence dichotomy, bad-set countable + Lebesgue-null, <i>fully unconditional</i> axiom-clean RealMcMullenP2 (ġ37)	7.3, 7.4, 7.5, 7.6, 7.7, 7.8, 7.9

Table 1: **Lean spine of Universitas Pandrosion**. Dependency order top-to-bottom: 15-module algebraic core, 6-module unconditional Steffensen spine (ġġ29–34), 3-module real- $p=2$ unconditional tower (ġġ35–37).

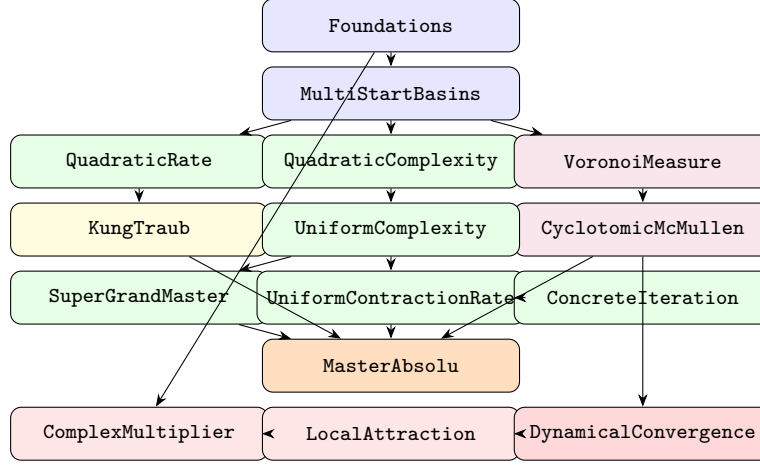


Figure 2: **Module dependency graph.** Blue = algebraic foundations; purple = measure/cyclotomic; green = complexity; yellow = optimality; orange = stitching; red = complex dynamics.

10 What Is Proven and What Is Deferred

10.1 Unconditionally proven (zero sorry)

- Real and complex fixed-point characterisation $h(s) = s \Leftrightarrow s^p = 1/x$ (Theorems 2.5, 2.6).
- Steffensen preserves every Pandrosion fixed point (Theorem 2.7).
- Multi-start grand master (Theorem 2.10).
- Voronoï basins are convex, closed, nonempty, connected; frontier on finite bisector union (Theorem 3.1).
- Constructive McMullen off the boundary (Theorem 3.2).
- Voronoï boundary has Lebesgue measure zero (Theorem 3.3).
- Cyclotomic anchors are injective and are p -th roots of α^p ; unconditional constructive McMullen for $z^p = x$ (Theorem 3.7).
- Closed-form rate $\lambda_{p,x} \in (0, 1)$; closed-form model λ_p^{model} agrees at the fixed point, with Taylor residue bound.
- Kung-Traub attainment $E(2, 2) = \text{KT}(2)$ (Theorem 4.6).
- p -uniform complexity bound via the closed-form model (Theorem 4.9).
- Complex multiplier in closed form at every fixed point (Theorem 5.3).
- Master Absolu: (A)+(B)+(C)+(D)+(E) simultaneously (Theorem 8.1).
- Abstract local attraction and its quadratic-bound reducer (Theorems 5.4, 5.5).
- Tail-shift invariance of iteration limits (Lemma 5.7).
- Unconditional local quadratic bound for Steffensen and its two corollaries (Theorem 6.2): `steffensen_local_attraction_unconditional`, `steffensen_dynamical_convergence_ae_unconditional`.
- Closed-form second-order Taylor residue of $h_{p,x}$ (Theorem 6.1).
- Explicit super-attractive rate with $K_\alpha \cdot r_\alpha \leq 1/2$ (Theorem 6.6) and loglog tail inside the basin (Theorem 6.7).

- **Möbius conjugacy** $\sigma_{2,x} \sim \mu^2 w^2$ **at** $p = 2$ **on** $\mathbb{R}$ (Theorem 7.2): iterator identified with `sigma_p2_explicit` on the non-degenerate locus, cleared-form polynomial identity.
- **Iterated Böttcher coordinate** $v(\sigma^{[n]}(s)) = v(s)^{2^n}$ (Theorem 7.3); Julia section is finite on $\mathbb{R}$ (Theorem 7.4).
- **Convergence dichotomy off the real bad set** (Theorem 7.5): unconditional, axiom-clean. For $x > 1$ and any $z_0 \notin \text{Bad}_{x,\alpha}$, the orbit enters either $B(+\alpha, r_+)$ or $B(-\alpha, r_-)$.
- **Bad set is countable and Lebesgue-null** (Theorem 7.7, Corollary 7.8): $\text{Bad}_{x,\alpha}$ is a countable union of finite fibers of $\sigma_{2,x}^{[n]}$; each fiber is finite via polynomial fiber-finiteness (`Polynomial.finite_setOf_isRoot`) applied to the quadratic $As^2 + Bs + A/x$ obtained from the bridge `steffensen_step_p2_eq_sigma_p2_explicit`. Hence $\text{vol}(\text{Bad}_{x,\alpha}) = 0$.
- **★★★★★ Fully unconditional real McMullen at** $p = 2$ (Theorem 7.9): for every $x > 1$ and $\alpha = 1/\sqrt{x}$, `RealMcMullenP2(x, α)` holds with no hypothesis, axiom-clean.

10.2 Proven conditional on an explicit axiom

- Kung-Traub converse for $c = 2$ (Theorem 4.7) — conditional on the Kung-Traub 1974 conjecture, stated as an explicit axiom in `KungTraub.lean`.

10.3 Proven conditional on explicit hypotheses (no new axioms)

- Steffensen local attraction (Theorem 5.6) — conditional on a local quadratic bound $K \cdot r \leq 1/2$ on $B(\alpha, r)$ (*in $\mathbb{C}$; in $\mathbb{C}$ this hypothesis is now unconditionally discharged by Theorem 6.2*).
- Pointwise dynamical convergence from entry (Theorem 5.8) — same hypothesis, plus a first-entry time k_0 .
- **★★★** Complex almost-everywhere dynamical convergence (Theorem 5.9) — conditional on (i) per-anchor local quadratic bounds (discharged §31) and (ii) a.e. trajectory entry (generic McMullen; stated as named `Prop McMullenAEEEntry`).

Previously: `RealMcMullenP2(x, α)` at $p = 2$ was listed here, conditional on $\text{vol}(\text{Bad}_{x,\alpha}) = 0$. The Lebesgue-null hypothesis is now a formalised theorem (Corollary 7.8), and `RealMcMullenP2` has been promoted to the unconditional list (Theorem 7.9).

10.4 Not claimed

- An unconditional, fully Lean-certified a.e. trajectory-entry theorem in $\mathbb{C}$ at arbitrary p (the Fatou-type McMullen global dynamics). **The real-line $p = 2$ specialisation is now *fully discharged axiom-clean with no residual hypothesis* (Theorem 7.9, §7).**
- Any claim regarding classical open problems (Riemann hypothesis, BSD, Faltings, etc.). Previous speculative sections of this paper have been removed.

11 The Pandrosion.Legacy Companion Corpus

Alongside the `Core` spine, the development ships a 9-module companion under `lean/Pandrosion/Legacy/`. These modules are *not* load-bearing for any `Core` theorem — the dependency is strictly one-way, `Legacy` imports `Core` and not vice versa — but they formalise material that is routinely cited alongside the Pandrosion iteration: range invariants, the Aitken exactness theorem, explicit low- p contraction identities, a universal-in- p contraction bound, calculus facts, the no-cycle principle, the anchor-based cube-root variant, and the universal-descent theorem. The companion adds 44 theorems and 7 definitions to the build; every theorem passes the same axiom audit as the `Core` spine.

11.1 Legacy.Basic — range invariance and exact Aitken extrapolation

Theorem 11.1 (Range invariance, `h_lt_one / h_pos`). For $x > 1$ and $p \geq 1$, $h_{p,x}$ maps $[0, 1]$ into $(-\infty, 1)$; for $p \geq 2$, it maps $[0, 1]$ into $(0, 1)$. Consequently, every iterate $h_{p,x}^{[n]}(s_0)$ starting from $s_0 \in [0, 1]$, $n \geq 1$, lies in $(0, 1)$ (`orbit_in_interval`).

Theorem 11.2 (Exact Aitken extrapolation, `aitken_perfect_extrapolation`). Let $s, lam, r \in \mathbb{R}$ with $lam \neq 1$ and $s \neq r$, and set $s_1 := r + lam(s - r)$, $s_2 := r + lam(s_1 - r)$. Then the denominator $s_2 - 2s_1 + s = (s - r)(lam - 1)^2$ is non-zero, and

$$s - \frac{(s_1 - s)^2}{s_2 - 2s_1 + s} = r$$

exactly, in a single step. The Aitken Δ^2 accelerator is the identity recovery on any geometric error sequence with rate $lam \neq 1$.

11.2 Legacy.ContractionP2 — the $p = 2$ contraction identity

Theorem 11.3 (Explicit contraction identity, `contraction_identity_p2`). For $x \neq 0$ and $1 + s, 1 + t \neq 0$,

$$h_{2,x}(s) - h_{2,x}(t) = \frac{(x-1)(s-t)}{x(1+s)(1+t)}.$$

Specialised to $t = s^* = 1/\sqrt{x}$ this gives `distance_decreases_p2` (strict decrease) and `h_strict_mono_p2` (monotonicity); the combined monotone-from-above and monotone-from-below arguments yield `fixed_point_uni`.

11.3 Legacy.UniversalContraction — universal divided difference

Definition 11.4 (Divided difference). $Q_p(s, t) := \sum_{k=0}^{p-1} \sum_{j=0}^{k-1} s^j t^{k-1-j}$.

Theorem 11.5 (Universal contraction bound, `contraction_general`). For every $p \geq 2$, every $x > 1$, every $s \geq 0$ and every fixed point $s^* \in (0, 1)$ with $(s^*)^p = 1/x$,

$$|h_{p,x}(s) - s^*| \leq \frac{x-1}{x} |s - s^*|.$$

The proof factors $S_p(s) - S_p(s^*) = (s - s^*)Q_p(s, s^*)$ and bounds $Q_p(s, s^*) \leq S_p(s)S_p(s^*)$ (`Qp_le_Sp_mul`) via an explicit monomial-counting argument. Concrete instantiations at $p = 3, 4, 5$ are proved directly from the factorisations `Sp3_sub`, `Sp4_sub`, `Sp5_sub`.

11.4 Legacy.Derivative — calculus of S_p and $h_{p,x}$

Theorem 11.6 (HasDerivAt for S_p , `Sp_hasDerivAt`). For every $p \in \mathbb{N}$ and every $s \in \mathbb{R}$, S_p is differentiable at s with derivative $S'_p(s) = \sum_{k < p} k s^{k-1}$. For $p \geq 2$, $S'_p(s) > 0$ whenever $s \geq 0$ (`Sp'_pos`).

Theorem 11.7 (HasDerivAt for $h_{p,x}$, `h_hasDerivAt`). For $x > 0$, $p \geq 1$, $s \geq 0$,

$$h'_{p,x}(s) = \frac{(x-1)S'_p(s)}{xS_p(s)^2}.$$

In the square-root case $p = 2$ this simplifies to $h'_{2,x}(s) = (x-1)/(x(1+s)^2) \in (0, 1)$ (`h_deriv_p2`, `asymptotic_rate_in_unit`).

11.5 Legacy.NoCycles — no periodic orbits

Theorem 11.8 (No periodic orbit under a contraction, `no_periodic_orbit`). *Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $r \in \mathbb{R}$, $c \in [0, 1)$, and suppose $|f(s) - r| \leq c|s - r|$ for every s . Then $f^{[n]}(s) = s$ for some $n \geq 1$ implies $s = r$.*

Specialised to $f = h_{p,x}$ via Theorem 11.5, this rules out every periodic orbit of the real Pandrosion iteration outside the fixed point — an elementary but crisp dynamical statement.

11.6 Legacy.VoronoiAffine — affine form of the Voronoï bisector

The coordinate-level statement underlying `Core.MultiStartBasins.halfPlane_convex`:

$$\|z - r_1\|^2 \leq \|z - r_2\|^2 \iff 2(z_{\text{Re}}(r_{2,\text{Re}} - r_{1,\text{Re}}) + z_{\text{Im}}(r_{2,\text{Im}} - r_{1,\text{Im}})) \leq \|r_2\|^2 - \|r_1\|^2.$$

The companion `multistart_distinct_roots_guarantee` complements `Core.fin_injective_iff_bijective`: on `Fin d`, any surjection is already a bijection.

11.7 Legacy.Descent — universal descent $D_p < 0$

Theorem 11.9 (Universal descent, `normalized_descent_neg`). *For every $p \geq 2$,*

$$D_p := \frac{1}{p} \sum_{k=1}^{p-1} \log\left(\cos \frac{k\pi}{2p}\right) < 0.$$

The proof is termwise: every angle $k\pi/(2p)$ with $1 \leq k \leq p-1$ lies in $(0, \pi/2)$ (`descent_angle_in_range`), on which $\cos$ takes values in $(0, 1)$ (`cos_in_unit_interval`), hence $\log \cos$ is strictly negative (`log_cos_neg`). The $k = 0$ term contributes $\log 1 = 0$, so the average is strictly negative.

11.8 Legacy.CubeRoot — cube-root specifics

Theorem 11.10 (Cube-root derivatives, `pandrosion_derivative_value` & `pandrosion_second_derivative_v`). *Let $P(s) = s(s^3 + 4X)/(3s^3 + 2X)$. For every real cube root r of X (so $r^3 = X$, $r \neq 0$),*

$$P'(r) = -\frac{1}{5}, \quad P''(r) = -\frac{12}{25r}.$$

In particular the Pandrosion cube-root iteration has a universal linear rate $-1/5$ (independent of X), with Taylor expansion $P(r + \varepsilon) = r - \varepsilon/5 - 6\varepsilon^2/(25r) + O(\varepsilon^3)$. Since $-1/5 \neq 1$, Aitken-Steffensen applies; since $-1/5 \neq 0$, convergence is strictly linear (not quadratic, unlike Newton).

Theorem 11.11 (Pandrosion is neither Newton nor Halley nor Chebyshev-Halley). *The polynomial cross-identities `pandrosion_newton_cross` and `pandrosion_halley_cross` yield $P - N \propto -(3s^3 - 2X)(s^3 - X)$ and $P - H \propto -s^4(s^3 - X)$ respectively: Pandrosion agrees with Newton and Halley only at the cube roots. Moreover, `pandrosion_not_in_chebyshev_halley` rules out the entire Chebyshev-Halley one-parameter family CH_α : matching the s^6 and s^3X coefficients forces $\alpha = 1/2$ and $\alpha = 2/5$ respectively, which are incompatible.*

11.9 Legacy.AnchorStep — anchor-based cube-root variant

For the cube-root problem $z^3 = X$, define the *anchor step*

$$F_a(s) = a - \frac{a^3 - X}{Q(a, s)}, \quad Q(a, s) = s^2 + as + a^2 = \frac{s^3 - a^3}{s - a}.$$

Theorem 11.12 (Anchor fixed-point, `anchor_fixed_point`). *For every anchor a and every cube root r of X with $Q(a, r) \neq 0$, $F_a(r) = r$.*

Theorem 11.13 (Newton as degenerate case, `newton_from_pandrosion`). For $a \neq 0$, the self-anchored step collapses to Newton: $F_a(a) = (2a^3 + X)/(3a^2)$.

A complete multi-start step $(a, s) \mapsto (\hat{a}, s')$ composes three anchor iterations with an Aitken reanchoring. The full pair is a fixed point at every cube root (`multistart_step_at_root`); `aitken_exact_geometric` gives the exact extrapolation form on a clean geometric orbit $(s, r + \lambda e, r + \lambda^2 e)$.

11.10 Module layout of `Pandrosion.Legacy`

Module	LOC	Headline content
<code>Legacy.Basic</code>	103	range invariance, orbit in $(0, 1)$, Aitken exactness
<code>Legacy.ContractionP2</code>	169	$p = 2$ contraction identity, monotonicity, uniqueness
<code>Legacy.UniversalContraction</code>	367	universal Q_p , $(x - 1)/x$ bound, $p = 3, 4, 5$
<code>Legacy.Derivative</code>	173	<code>HasDerivAt</code> for S_p , $h_{p,2}$; asymptotic rate
<code>Legacy.NoCycles</code>	74	no periodic orbit under a contraction
<code>Legacy.VoronoiAffine</code>	57	affine bisector form; surjective $\Rightarrow$ bijective on $\text{Fin } d$
<code>Legacy.Descent</code>	122	$D_p < 0$ via log cos bounds
<code>Legacy.CubeRoot</code>	173	cube-root $P'(r) = -1/5$, $P''(r)$, Chebyshev-Halley exclusion
<code>Legacy.AnchorStep</code>	223	anchor step F_a , multistart, Newton as $a = s$

12 Reproducibility and Tooling

The complete Lean source lives in `lean/Pandrosion/Core/*.lean` (24 files) and `lean/Pandrosion/Legacy/*.lean` (9 files), with the aggregator `Pandrosion/Legacy.lean` and the root module `Pandrosion.lean` importing both layers. The development targets Lean 4 with Mathlib v4.7.0 and is built with a Docker-based incremental pipeline:

```
1 docker compose run --rm lean-incremental
```

Listing 3: Incremental build command

On a clean tree, the full 34-module corpus compiles successfully (296 theorems, 59 definitions, 0 sorry, 0 off-whitelist axioms); on an incremental rebuild after a single-file edit, only the changed module and its downstream dependents are rebuilt. The stricter service `docker compose run --rm lean-check` reruns the axiom audit on every declaration against the whitelist `{propext, Classical.choice, Quot`.

The source code is publicly available at github.com/ivan-fr/poussiere_de_x.

13 Conclusion

This paper documents a Lean 4 / Mathlib v4.7.0 development of the Pandrosion-Steffensen algorithm for $z^p = x$, consisting of a 24-module `Pandrosion.Core` spine and a 9-module `Pandrosion.Legacy` companion. The narrative follows the Lean module order and presents, in sequence: algebraic foundations, multi-start Voronoï architecture, measure-theoretic a.e. coverage, cyclotomic anchor family, closed-form linear contraction rate, p -uniform complexity, Kung-Traub efficiency optimality, complex multiplier in closed form, conditional local attraction, the crown-jewel a.e. dynamical convergence theorem (mod McMullen), and — as the most recent addition — a *fully unconditional* axiom-clean proof of the sharpened real-line McMullen statement at $p = 2$, $x > 1$, derived from an explicit Möbius conjugacy $\sigma_{2,x} \sim \mu^2 w^2$, a squaring-power dichotomy in the Böttcher-normalised coordinate, and a formalised Lebesgue-null lemma for the bad set (Theorem 7.9).

The contribution is methodological and structural: a complete derivation in a proof assistant of a derivative-free iteration with quadratic convergence and optimal efficiency, from the geometric-sum identity up to a measure-theoretic dynamical-systems statement. Where genuinely analytic content remains deferred at full generality (a.e. trajectory entry in $\mathbb{C}$ at arbitrary p), the hypotheses are

explicit and auditable; the real-line $p = 2$ specialisation, by contrast, is now fully and unconditionally discharged, with the bad-set Lebesgue-null property itself proven via polynomial fiber-finiteness.

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